

DEMESHWAR RANA

AI & Data Science Student | Machine Learning & Deep Learning

Personal Information

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Professional Summary

MSc Data Science & AI student at Gisma University (Potsdam) specialising in machine learning and deep neural networks (CNN, RNN/GRU/LSTM, transfer learning). Hands-on experience training and rigorously validating models with TensorFlow/Keras and Scikit-learn, from raw timeseries and image data through to a result that holds up under proper testing. Seeking a Werkstudent role in Data Science, Machine Learning, or AI Engineering.

I am eager to bring this experience in machine learning and rigorous model validation to a team working on real, data-driven problems.

Primary stack: Python, TensorFlow/Keras, CNN, RNN/GRU/LSTM, Transfer Learning, Scikit-learn

Work History

01/2024 –
05/2024

Robotics & Data Intern

Centre of AI and Robotics – IIT Mandi | Mandi, India

Tech stack: SolidWorks, Siemens NX (Unigraphics), Motion & Stress Analysis

- Identified and resolved a structural weak point in a quadruped robot frame through motion and stress analysis, catching the issue before fabrication and avoiding a costly re-design
- Engineered a complete cleaning assembly for a ship hull robot using mechanism inversion to achieve precise 45° brush oscillations, enabling reliable automated hull cleaning without manual rework

Academic
Project

Day-Ahead Electricity Demand Forecasting

National Grid ESO, GB | Self-directed coursework project

Tech stack: Python, TensorFlow/Keras (SimpleRNN, GRU, LSTM), Pandas, Scikit-learn, Plotly

- Benchmarked SimpleRNN, GRU, and LSTM architectures under identical conditions (fixed seed, Dropout, EarlyStopping, time-ordered 80/10/10 split); selected GRU as the final model
- Sourced 16 years of half-hourly ESO data but trained on the post-2022 window only, avoiding a temporal distribution shift from the full historical span
- Achieved a test R^2 of 0.87 (MAE 1,708 MW), consistently outperforming a naive persistence baseline (R^2 0.79)

Academic
Project

Image Classification: CNN & Transfer Learning (EfficientNetB0)

Tech stack: Python, TensorFlow/Keras, CNN, EfficientNetB0, Matplotlib

- Built a regularised CNN classification pipeline on CIFAR-10 using Dropout, BatchNorm, and data augmentation to control overfitting
- Applied transfer learning with a pretrained EfficientNetB0, improving test accuracy meaningfully over the from-scratch CNN baseline while also reducing total training time

Education

01/2026 – 01/2028 (expected)	M.Sc. in AI, Data Science & Digital Business Gisma University of Applied Science Potsdam, Germany Key modules: Machine Learning, Neural Networks, Business Analytics, Data-Driven Decision Making, Digital Business Strategy
08/2021 – 06/2024	B.Tech in Mechanical Engineering Jawaharlal Nehru Government Engineering College Sundernagar, India
2018 – 2021	Diploma in Mechanical Engineering Government Polytechnic College Sundernagar, India

Additional Skills & Interests

Languages

- English (C1), German (A2, actively improving), Hindi (Native)

Interests

- I love building neural network projects from scratch, and my favorite so far was training a GRU model to forecast a full day of national electricity demand.
- I enjoy exploring open-ended data problems that do not have a clean textbook answer, since that is where the real learning happens.